MAGNETOM

Troubleshooting Guide System

Cooling

Aera
Skyra
Avanto fit
Skyra fit
Prisma
Prisma fit

Document Version

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1.1 Notes

Do not use any connection to a drinking water system for water filling!

1.1.1 General Remarks

In this section, you will find the strategy and test procedures for cooling troubleshooting.

The main tool for troubleshooting consists of error messages delivered from the SEP, IFP and chiller and shown in the error log file of the Service platform.

Please refer to the KKT User Manual for chiller troubleshooting.

2.1 Error Codes of the SEP Controller

Code[1]	Description	Explanation
ECO	ECO mode active	
CAL	Calibrate mixing valve (only at start-up)	
88.8	Primary water temperature	
F01	Overload MREF compressor	Warning - compressor motor protection switch
F02	Low water flow MBB	Low water flow MBB circuit
F03	Low water flow CBB	Low water flow CBB circuit
F04	Low water flow primary water circuit	Primary water flow low
F05	Overload water pump	
F06	Inverter error	
F07	Phase sequence error	
F11	Temp < 16.0°C	Secondary water temperature low warning
F12	Temp > 25.0°C	Secondary water temperature high warning
F14	Temp < 10.0°C	Secondary water temperature low error
F15	Temp > 30.0°C	Secondary water temperature high error
F16	Temp > 35.0°C	Secondary water temperature high error stop
F17	Sensor error	Secondary temperature sensor error
F21	Temp < 4.0°C	Primary water temperature low warning
F22	Temp > 14.0°C	Primary water temperature high warning
F24	Temp < 1.0°C	Primary water temperature low error
F25	Temp > 25.0°C	Primary water temperature high error
F27	Sensor error	Primary temperature sensor error
F31	Pressure > 5.8 bar	Supply water pressure high warning
F34	Pressure > 6.0 bar	Supply water pressure high error
F37	Sensor error	Supply pressure sensor error

Code[1]	Description	Explanation
F41	Pressure < 0.7 bar	Return pressure low warning
F42	Pressure > 1.3 bar	Return pressure high warning
F44	Pressure < 0.3 bar	Return pressure low error
F45	Pressure > 1.8 bar	Return pressure high error
F47	Sensor error	Return pressure sensor error
F51		Inverter control low warning
F52		Inverter control high warning
F54		Inverter control low error
F55		Inverter control high error

1. Code is displayed on the controller board

2.2 Chiller

Symptom	Event Log	Probable Cause	Service Action
▪ IFPK2300 stopped sending heart beats. ▪ IFPK2300 sent out a bootup message	MRI_PFW 260 IFPK2300 error: The component was reset unexpectedly	SD memory card of the Chiller is defect	▪ Replace the SD card ▪ Reboot Meas&Recon ▪ Switch off/on the MR system

2.2.1 Chiller / IFP run and fault messages

Run and fault messages are displayed on the IFP board (→ 1/Fig. 1 Page 9) or the chiller board.

Tab. 1 Run and fault messages

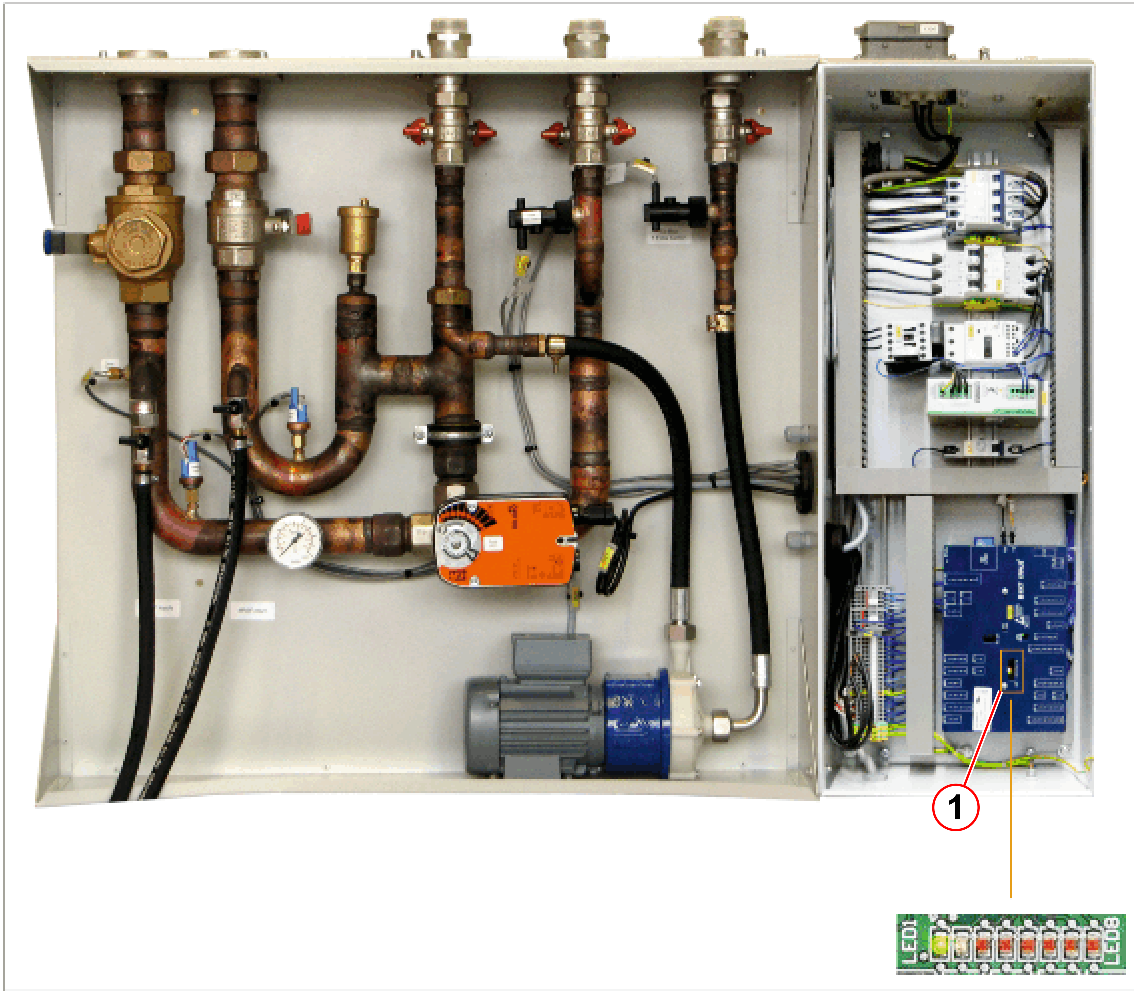
1	2	3	4	5	6	7	8	Description
x	x	x	x	x	x	x	x	LAMP TEST/RESET / Activated when the reset button is pressed for 1 second.
		x	x					Emergency stop (open circuit)
							x	Refrigerant high pressure warning
						x		Refrigerant low pressure warning
		x				x	x	Water temp return high / could be activated during startup or after power down
		x			x			Superheat <3K
		x			x		x	Temp. sensor suction gas (refrigerant side)
		x			x	x		Superheat >15K
		x			x	x	x	Water temp supply high
		x		x				Low pressure chiller (refrigerant)
		x		x			x	Sensor booster water
		x		x		x		Sensor free cooling unit water
		x		x		x	x	Water return pressure sensor broken. Chiller AI1
		x		x	x			Refrigerant high pressure (error/AI7)
		x		x	x		x	Refrigerant low pressure (error/AI8)
		x		x	x	x		Free cooling unit temperature sensor malfunction
		x		x	x	x	x	Water return pressure, pump stop (0.1 bar)

1	2	3	4	5	6	7	8	Description
		x	x	x				Phase supervision
		x	x	x			x	Flow error protection ~ 1 bar pressure difference
		x	x	x		x	x	Thermal contact FAN 2
		x	x	x	x			Frequency drive pump
		x	x	x	x		x	Thermal contact free cooling unit fan
		x	x	x	x	x		Frequency drive compressor 1
		x	x	x	x	x	x	Frequency drive FAN 1
		x	x			x		High pressure chiller refrigerant
		x	x			x	x	Frequency drive booster FAN
		x	x		x			Motor protection compressor 2
		x	x		x		x	Motor protection compressor booster
		x	x		x	x		Motor protection FAN 2
		x	x		x	x	x	Motor protection FCU
	x[1]							Service required / LED is blinking
	x							Service mode / LED is ON
	x						x	Pipes connected wrong or sensors on IFP changed
	x					x		40°C safety cutoff for the pump
	x					x	x	Water supply pressure high
	x				x			Sensor temp return error chiller
	x				x		x	Water temp supply low
	x				x	x		Sensor temp supply error IFP
	x				x	x	x	Return pressure LOW
	x			x				Flow volume MBB
				x			x	Chiller communication / POF-PCF
				x		x		SD card removed
	x			x	x			Flow volume CBB
	x			x	x		x	Auxiliary switch MREF
	x			x	x	x		Ambient high
	x			x	x	x	x	Sensor temp supply error chiller
			x					IFP supply pressure AI2

1	2	3	4	5	6	7	8	Description
			x	x				IFP return pressure AI3
			x	x			x	Ambient temperature AI10
			x	x		x		Refrigerant high pressure sensor booster AI9
	x		x	x	x	x	x	Supply pressure high
			x				x	Motor protection pump 1 TX box
			x			x		Motor protection pump 2 TX box (if installed)
			x			x	x	Flow switch pump 1 TX box
			x		x			Flow switch pump 2 TX box (if installed)

1. LED is blinking

Fig. 1: IFP PCB



(1) 8-bit LEDs

2.3 CAN24_7 Interface

If the MR system is switched off, Magnet and Cooling parameters can be read out via the CAN24_7 interface.

- Log in via RDIAG.
- Select Update.
- Press submit button to perform a helium measurement and read out the parameter.

Fig. 2: CAN 24_7 log in window

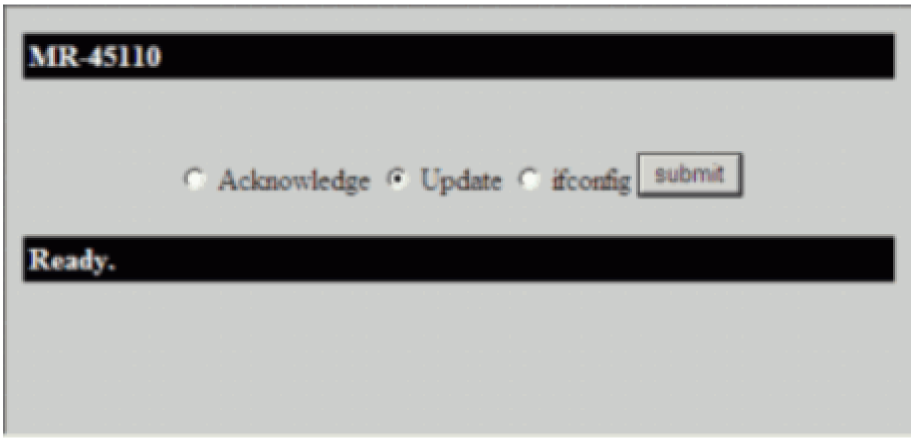
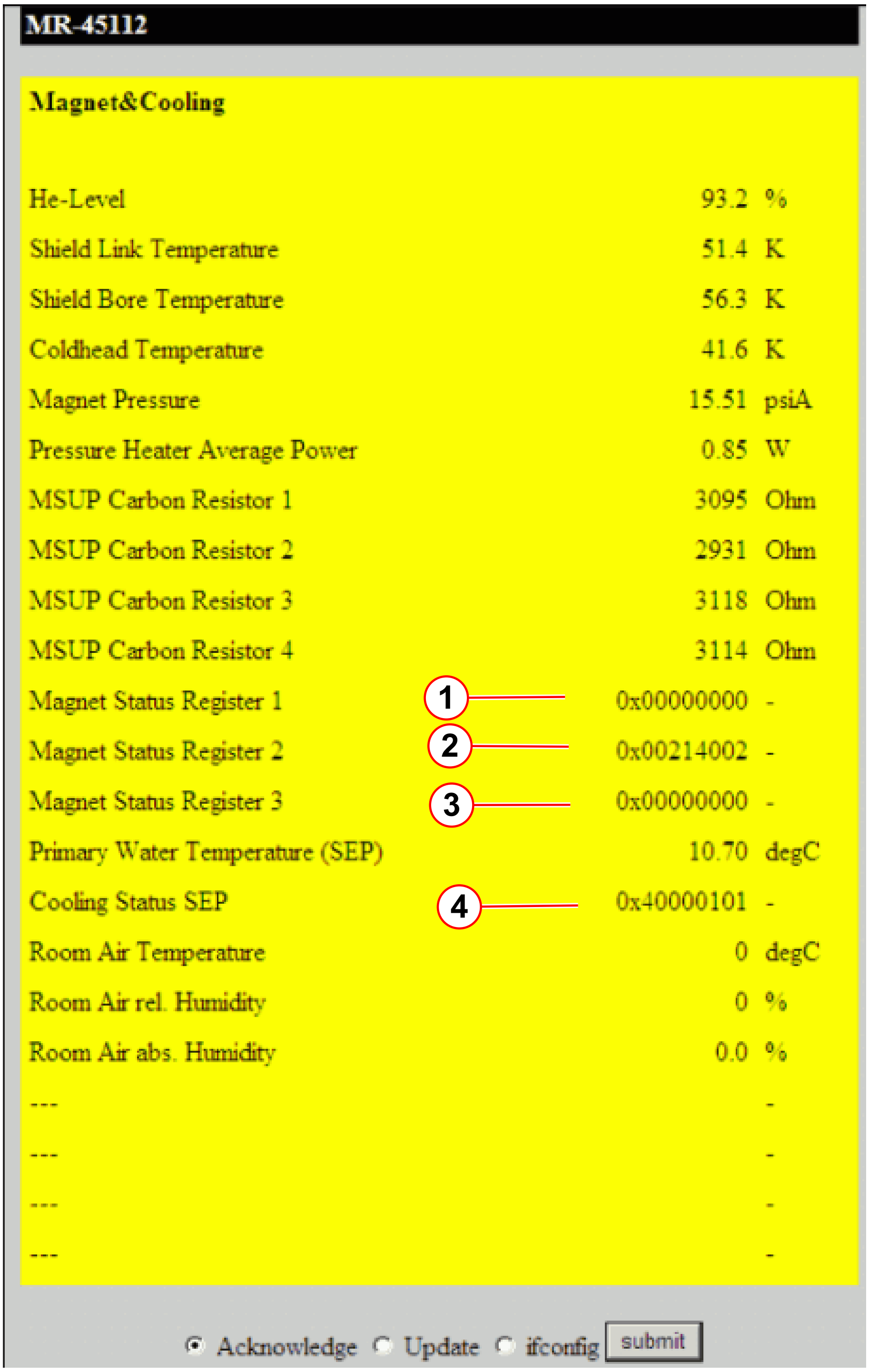


Fig. 3: CAN24_7 interface



(1) Magnet Status Register 1
(2) Magnet Status Register 2
(3) Magnet Status Register 3
(4) SEP Status Register

Tab. 2 SEP Status register

Cooling Status SEP	Hex code, (4/Fig. 3 Page 11)	40000101
	Binary code	01000000000000000000000100000001
Status Bit	**Description**	**comment**
0	SystemType	0=XJ, 1=XQ
1	SecPumpRunning	0=on, 1=off
2	FlowCBB	0=ok, 1=Fault
3	FlowMBB	0=ok, 1=Fault
4	PrimFlow	0=ok, 1=Fault
5	MotProtSwitchMREF	0=ok, 1=Fault
6	MotProtSwitchPump	0=ok, 1=Fault
7	InverterError	0=ok, 1=Fault
8	SecondaryTempHighWarning	0=ok, 1=Fault
9	SecondaryTempLowWarning	0=ok, 1=Fault
10	PrimaryTempHighWarning	0=ok, 1=Fault
11	PrimaryTempLowWarning	0=ok, 1=Fault
12	SupplyPressureHighWarning	0=ok, 1=Fault
13	ReturnPressureLowWarning	0=ok, 1=Fault
14	InverterControlHighWarning	0=ok, 1=Fault
15	InverterControlLowWarning	0=ok, 1=Fault
16	SecondaryTempHighError	0=ok, 1=Fault
17	SecondaryTempHighStop	0=ok, 1=Fault
18	SecondaryTempLowError	0=ok, 1=Fault
19	PrimTempHighError	0=ok, 1=Fault
20	PrimTempLowError	0=ok, 1=Fault
21	SupplyPressureHighError	0=ok, 1=Fault
22	ReturnPressureLowError	0=ok, 1=Fault
23	ReturnPressureHighWarning	0=ok, 1=Fault
24	InverterControlHighError	0=ok, 1=Fault
25	InverterControlLowError	0=ok, 1=Fault

Cooling Status SEP	Hex code, (4/Fig. 3 Page 11)	40000101
	Binary code	01000000000000000000000100000001
Status Bit	Description	comment
26	SecTempSensorBroken	0=ok, 1=Fault
27	PrimTempSensorBroken	0=ok, 1=Fault
28	SupplyPressureSensorBroken	0=ok, 1=Fault
29	ReturnPressureSensorBroken	0=ok, 1=Fault
30	ECO_mode	0=off, 1=on
31	PhaseSupervision	0=ok, 1=Fault

Tab. 3 Magnet Status register1

Magnet Status Register1	Hex code: (1/Fig. 3 Page 11)	00000000
	Binary code	00000000000000000000000000000000
Status Bit	Description	comment
0	TurretTemperatureAlarm	0=ok, 1=Fault
1	TurretTemperatureWarning	0=ok, 1=Fault
2	ColdheadTemperatureAlarm	0=ok, 1=Fault
3	ColdHeadTemperatureWarning	0=ok, 1=Fault
4	Shield1LinkAlarm	0=ok, 1=Fault
5	Shield1LinkWarning	0=ok, 1=Fault
6	Shield1BoreAlarm	0=ok, 1=Fault
7	Shield1BoreWarning	0=ok, 1=Fault
8	Shield2LinkAlarm	0=ok, 1=Fault
9	Shield2LinkWarning	0=ok, 1=Fault
10	Shield2LinkWarning	0=ok, 1=Fault
11	Shield2BoreWarning	0=ok, 1=Fault
12	CompressorPressure	0=ok, 1=Fault
13	CompressorOverTemperature	0=ok, 1=Fault

Magnet Status Register1	Hex code: (1/Fig. 3 Page 11)	00000000
	Binary code	00000000000000000000000000000000
Status Bit	Description	comment
14	CompressorMotorError	0=disabled, 1=enabled
15	CompressorGasFault	0=ok, 1=Fault
16	CompressorWaterFault	0=ok, 1=Fault
17	reserved	0=ok, 1=Fault
18	CompressorPowerError	0=ok, 1=Fault
19	reserved	
20	reserved	
21	reserved	
22	reserved	
23	reserved	
24	reserved	
25	reserved	
26	reserved	
27	reserved	
28	reserved	
29	reserved	
30	reserved	
31	reserved	

Tab. 4 Magnet Status register2

Magnet Status Register2	Hex code: (2/Fig. 3 Page 11)	00214002
	Binary code	00000000001000010100000000000010
Status bit	Description	comment
0	SupervisoryTest	0=test inactive, 1=test active
1	ZeroField	0=no field, 1=field present

Magnet Status Register2	Hex code: (2/Fig. 3 Page 11)	00214002
	Binary code	00000000001000010100000000000010
Status bit	Description	comment
2	HeliumAlarm	0=ok, 1=Fault
3	HeliumWarning	0=ok, 1=Fault
4	HeliumProbeReference	0=ok, 1=Fault
5	HeliumProbe1Status	0=ok, 1=Fault
6	HeliumProbe2Status	0=ok, 1=Fault
7	EisDriveSignalAlarm	0=ok, 1=Fault
8	HeliumLevelTestStatus	0=off, 1=on
9	EisTestStatus	0=off, 1=on
10	EisDecayTest	0=off, 1=on
11	PressureHeaterStatus	0=off, 1=on
12	MonthEventLogStatus	0=no change, 1=change
13	ColdheadValueLogStatus	0=no change, 1=change
14	MonthValueLogStatus	0=no change, 1=change
15	LvqdLogStatus	0=Not Triggered, 1=Triggered
16	StatusLogStatus	0=no change, 1=change
17	EepromParameterAlarm	0=ok, 1=Fault
18	SerialAdcReferenceStatus	0=ok, 1=Fault
19	EepromCcr	0=ok, 1=Fault
20	IbuttonStatus	0=ok, 1=Fault
21	CompressorStatus	0=off, 1=on
22	CompStaticPressTestStatus	0=off, 1=on
23	MPSActive	0=Not Active, 1=Active
24	MagnetQuench	0=Field, 1=Quench
25	CompressorAlarmStatus	0=ok, 1=Fault
26	AlarmBoxStatus	0=ok, 1=Fault
27	ABXconfig	0=ok, 1=Fault

Magnet Status Register2	Hex code: (2/Fig. 3 Page 11)	00214002
	Binary code	00000000001000010100000000000010
Status bit	Description	comment
28	LVQDDetecStatus	"0=ok, 1=LVQD Event Detected"
29	RampQuenchStatus	0=No Ramp Quench, 1 = Ramp Quench
30	reserved	
31	reserved	

Tab. 5 Magnet Status register3

Magnet Status Register3	Hex code: (3/Fig. 3 Page 11)	00000000
	Binary code	00000000000000000000000000000000
Status bit	Description	comment
0	QuenchlogStatus	0=Logging, 1=Quench Log File Recorded
1	ErduOperated	0=off, 1=on
2	reserved	
3	ERDUTestRunningStatus	0=ok, 1=Fault
4	ERDUButton1Status	0=ok, 1=o/c, 2= s/c
5	ERDUButton1Activated	0=ok, 1=Fault
6	ERDUButton2Status	0=ok, 1=o/c, 2= s/c
7	ERDUButton2Activated	0=ok, 1=Fault
8	ERDUButton3Status	0=ok, 1=o/c, 2= s/c
9	ERDUButton3Activated	0=ok, 1=Fault
10	ErduBatteryAlarm	0=ok, 1=Fault
11	ErduBatteryWarning	0=ok, 1=Fault
12	SwitchHeaterStatus	0=off, 1=on
13	SwitchHeaterAlarm	0=ok, 1=Fault
14	SwitchHeaterDriveAlarm	0=ok, 1=Fault
15	QuenchHeaterAlarm	0=ok, 1=Fault

Magnet Status Register3	Hex code: (3/Fig. 3 Page 11)	00000000
	Binary code	00000000000000000000000000000000
Status bit	Description	comment
16	VentFault	0=ok, 1=Fault
17	PressureBuildTestStatus	0=off, 1=on
18	PressureBuildTestResult	0=ok, 1=Fault
19	PressureDriveAlarm	0=ok, 1=Fault
20	MagnetPressureHighStatus	0=ok, 1=Fault
21	MagnetLowPressureStatus	0=ok, 1=Fault
22	AverageHeaterPowerStatus	0=ok, 1=Fault
23	PressureSwitch	0=ok, 1=Fault
24	P36VStatus	0=ok, 1=Fault
25	P15VStatus	0=ok, 1=Fault
26	P5VStatus	0=ok, 1=Fault
27	5VIsoStatus	0=ok, 1=Fault
28	3V3Status	0=ok, 1=Fault
29	PressureManualStatus	0=OFF, 1=ON
30	reserved	
31	reserved	

3.1 SEP

3.1.1 Drain procedure of the SEP (secondary water circuit)

3.1.1.1 Required tools

- Filling adapter (part of system delivery)
- Bucket
- Water hose (part of system delivery)

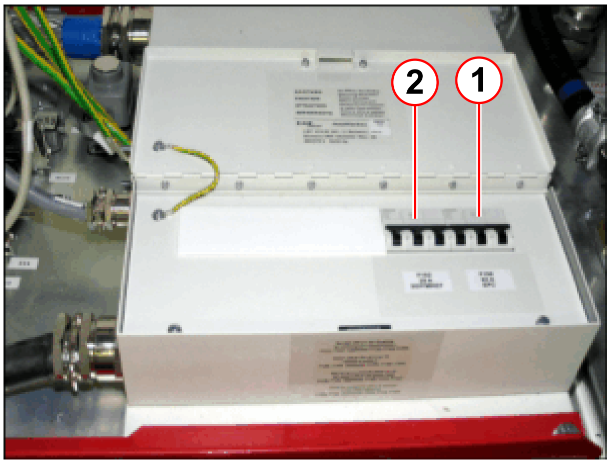
3.1.1.2 Procedure

- Switch off the compressor unit
- Switch off circuit breaker EPC / F102

Fig. 4: SEP

(1) Main power switch

Fig. 5: Circuit breakers in EPC mains box

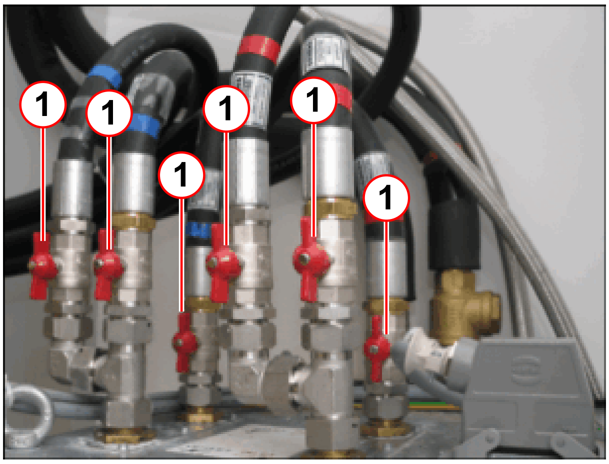


(1) F100 63A EPC
(2) F102 25A SEP/IFP - MREF

Drain the complete secondary water circuit of the SEP

- Close the shutoff valves on top of the SEP (➔ 1/Fig. 6 Page 19)

Fig. 6: SEP water connection



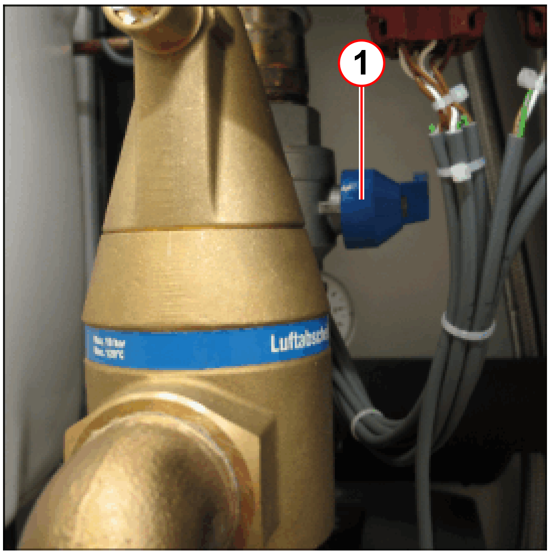
(1) Shutoff valve

Drain the secondary water circuit of the SEP for component exchange

- Close the ball valves (➔ Fig. 7 Page 19), (➔ 1/Fig. 8 Page 19).

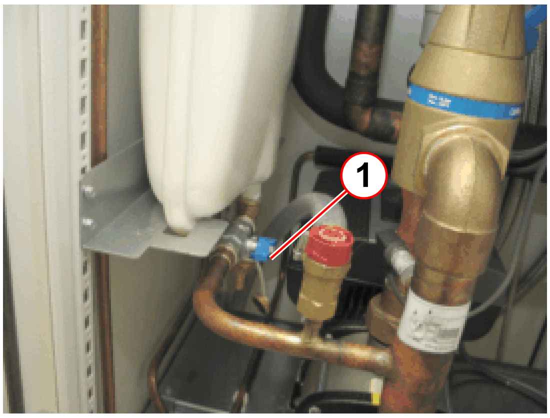
Fig. 7: Ball valve with strainer

Fig. 8: SEP



(1) Ball valve

Fig. 9: Expansion vessel

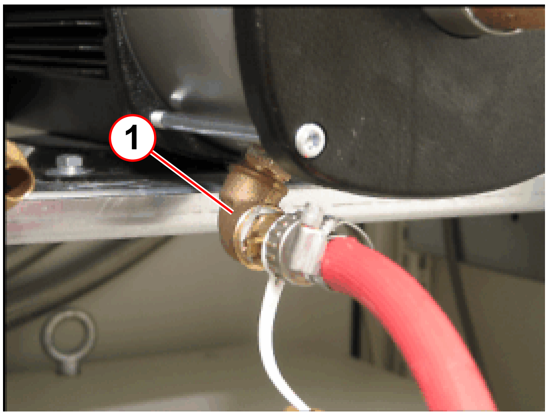


(1) Shut-off valve

- Close the shut-off valve on the expansion vessel.

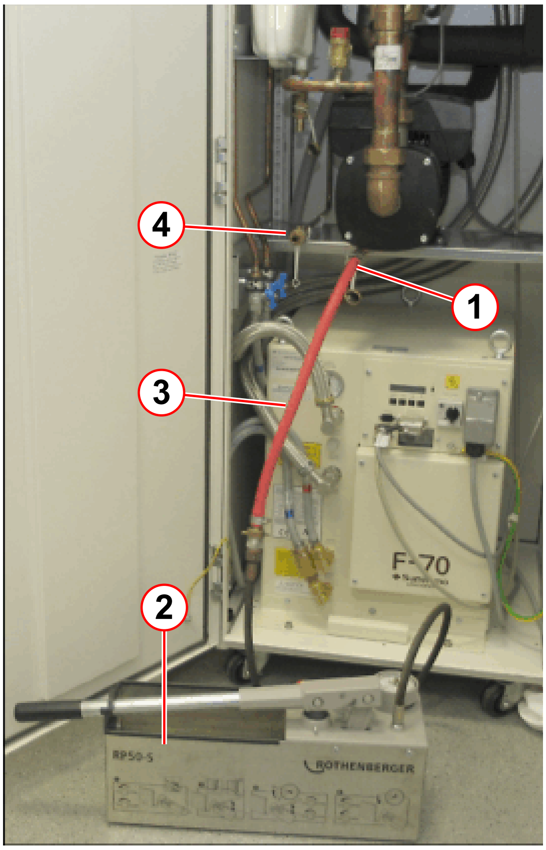
- Connect the water hose to the drain and fill port of the SEP.
- Run the water hose to a drain or a container.

Fig. 10: Drain / fill port



(1) Drain / fill port secondary circuit

Fig. 11: SEP

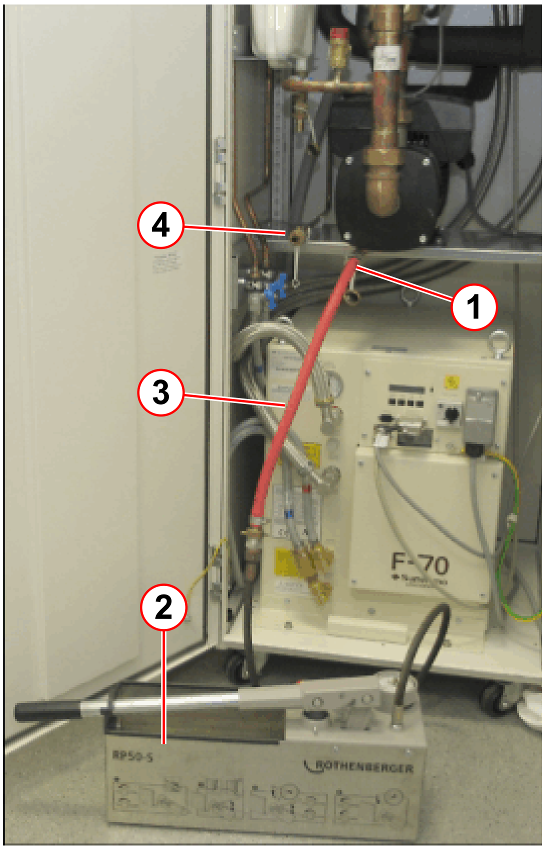


(1) Drain / fill port secondary circuit
(2) Water pump
(3) Water hose
(4) Drain / fill port primary circuit

Fig. 12: Drain / fill port

- Open the drain port using the cap of the drain valve.
- Loosen the screw of the valve above the air bleeder a little. Air entering the system will help drain the SEP.

Fig. 13: SEP



(1) Drain / fill port secondary circuit
(2) Water pump
(3) Water hose
(4) Drain / fill port primary circuit

3.1.2 Filling the SEP

> ℹ Do not use any connection to a drinking water system.

3.1.2.1 Required tools

- Filling adapter (part of system delivery)
- Water hose (part of system delivery)
- Water pump (if no on-site water supply is available, manual water pump part # 8115136)

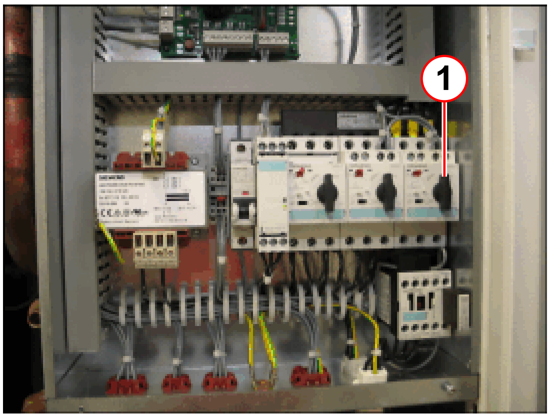
3.1.2.2 Procedure: Filling the water circuit

- Switch off the compressor unit using the main power switch.
- Switch off the water pump using Q3.

Fig. 14: SEP

(1) Main power switch

Fig. 15: SEP electronic cabinet



(1) Q3 water pump

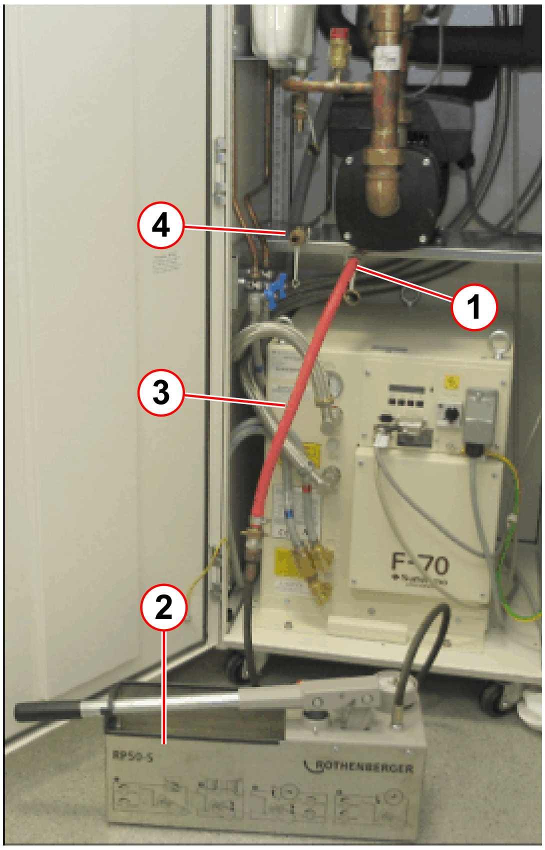
- Connect the water hose from the on-site water supply to the filling port of the SEP

  or

  connect the water hose from the water pump to the filling port of the SEP

> Before connecting the hose to the SEP, the hose should be filled with water and not contain air.

Fig. 16: SEP



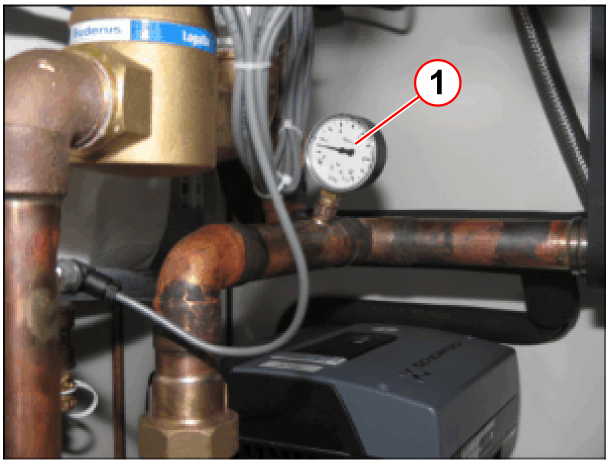
(1) Drain / fill port secondary circuit
(2) Water pump
(3) Water hose
(4) Drain / fill port primary circuit

Fig. 17: Drain / fill port

- Open the drain/fill port valve using the back of the cap
- Pump water into the circuit.
- Close the valve when the pressure manometer indicates the correct pressure:

	SEP pressure manometer
Static pressure	1.0 $_{-0.5}$ bar

Fig. 18: SEP water pressure



(1) Pressure manometer

Fig. 19: Pressure manometer

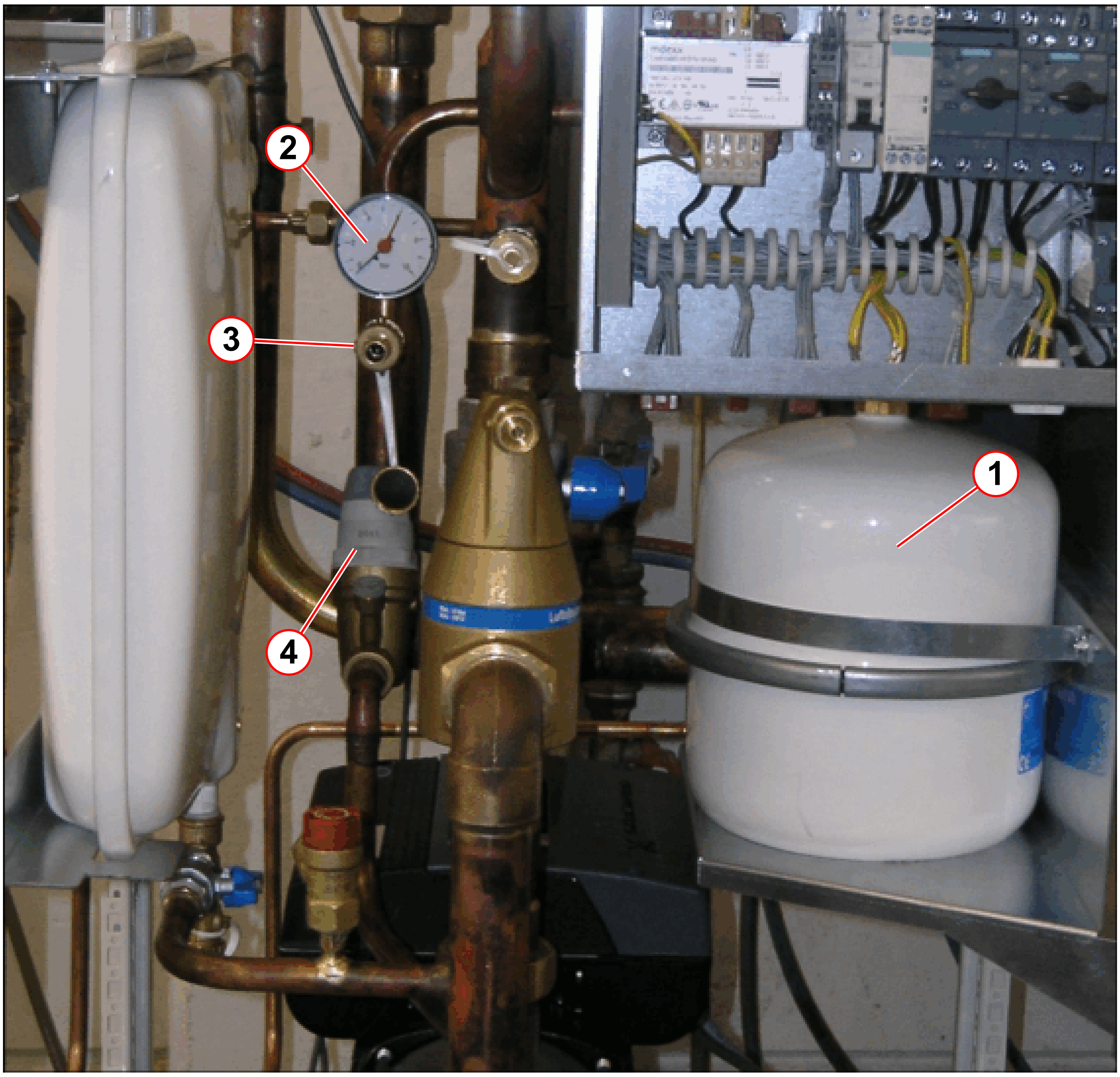
- Switch on the water pump Q3.
- Switch on the compressor unit.
- Check the system for leakages.
- Remove the filling hose.
- Close the filling port with the blanking plug.

3.1.3 Filling the auto-refill device.

This description is valid for a SEP with an auto-refill device.

> i
>
> The auto-refill device maintains the secondary water return pressure constant at 1.0 bar.
>
> Water is refilled automatically if water loss appears.
>
> Prerequisite: The auto-refill device is filled with water and the pressure regulator is adjusted to 1.0 bar.

Fig. 20: Auto-refill device



(1) Expansion vessel
(2) Pressure manometer
(3) Fill port of auto-refill device
(4) Pressure regulator

- Connect a water hose to the fill port of the auto-refill device. (→ 3/Fig. 20 Page 26)
- Open the fill port slowly.
- Fill the auto-refill device to a range between 4.0 bar and 6.0 bar.
- Check the pressure via the pressure manomer.(→ 2/Fig. 20 Page 26)

i The expansion vessel needs a filling if the pressure at the pressure manometer is below 1 bar.

3.1.4 Cleaning the Strainer

3.1.4.1 Required tools

- 50 mm open-end wrench
- Needle nose pliers
- Filling adapter (part of system delivery)
- Water hose (part of system delivery)
- Water pump (if no on-site water supply is available, manual water pump material number 8115136)
- Bucket (cup)

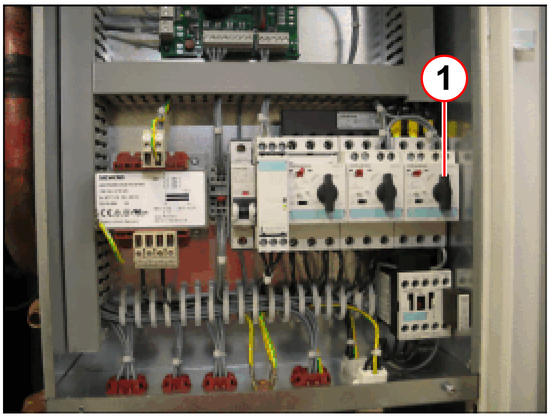
3.1.4.2 Procedure

- Switch off the compressor unit using the main power switch. (➔ 1/Fig. 21 Page 27)
- Switch off the water pump using Q3. (➔ 1/Fig. 22 Page 27)

Fig. 21: SEP

(1) Main power switch

Fig. 22: SEP electronic cabinet



(1) Q3 water pump

- Slowly close the ball valve. (➔ Fig. 24 Page 28)

Fig. 23: SEP - ball valve open

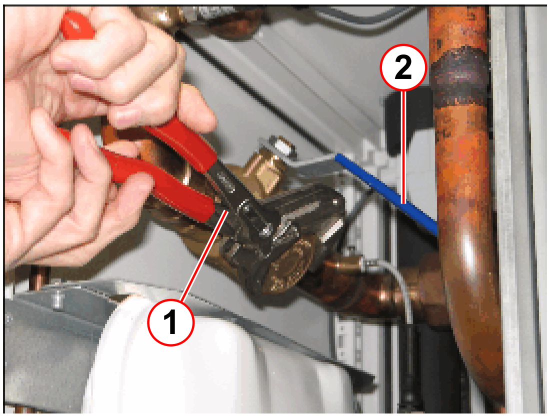
Fig. 24: SEP - ball valve closed

- Remove the valve cap slowly by using an open-end wrench.

> The closed ball valve contains approximately 200 ml of cooling water. Use a cup to collect the water.

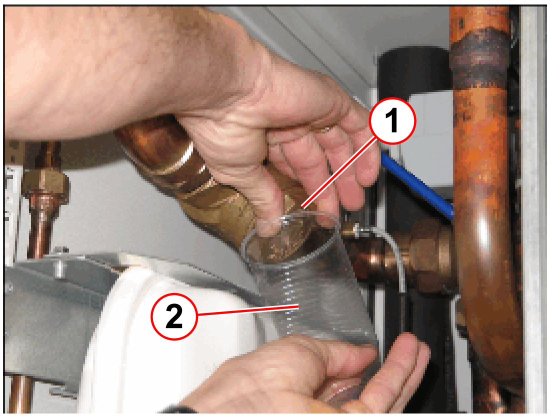
- Use a cup to collect the water. (➔ 2/Fig. 26 Page 28)

Fig. 25: SEP - ball valve with strainer



(1) Open-end wrench
(2) Ball valve - closed

Fig. 26: SEP - ball valve

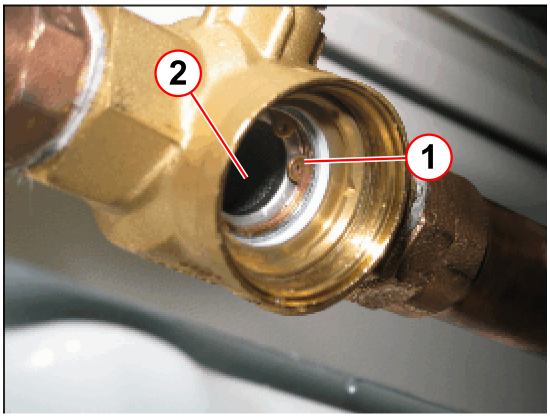


(1) Valve cap
(2) Cup

- Remove the circlip using needle nose pliers. (➔ 1/Fig. 27 Page 29)

- Remove the strainer.

Fig. 27: SEP - strainer



(1) Circlip
(2) Strainer

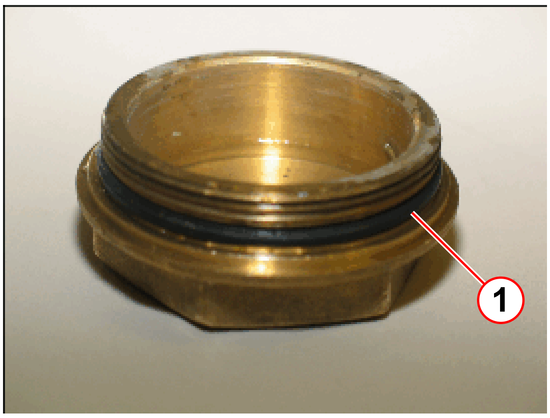
Fig. 28: Strainer

- Inspect the strainer for contamination and clean it by rinsing under tap water.
- Open the drain/fill port valve using the back of the cap
- Pump water into the circuit.
- Close the valve when the pressure manometer indicates the correct pressure:

Reassemble the strainer

- Re-install the strainer.
- Mount the strainer with the circlip.
- Check the O-ring of the valve cap.

Fig. 29: Ball valve cap



(1) O-ring

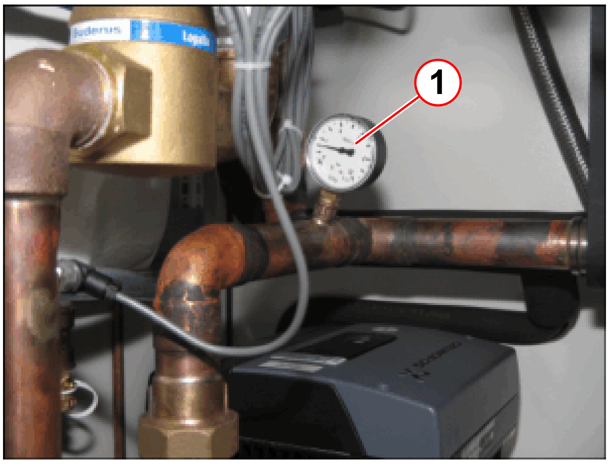
- Screw the valve cap into the ball valve.
- Slowly open the ball valve.
- Check for leaks on the valve cap.

Final steps

- Check the water pressure
    » Refill water if pressure < 1 bar

	SEP pressure manometer
Static pressure	$1.0_{-0.5}$ bar

Fig. 30: SEP water pressure



(1) Pressure manometer

Fig. 31: Pressure manometer

- Switch on the water pump Q3
- Switch on the compressor unit.
- Check for leaks.

3.1.5 Testing the expansion vessel

> Before replacing the pump because of pressure problems, test the pressure of the expansion vessel.

3.1.5.1 Required Tools

- Air pump with auto-valve delivered with the system (→ Fig. 33 Page 31).
- Measuring device: Manometer part #3055514 (→ Fig. 32 Page 31).
- Standard tool set.
- Water pump (manual water pump, part #8115136), (→ 2/Fig. 16 Page 24).
- Water hoses (→ 3/Fig. 16 Page 24).

Fig. 32: Measuring device

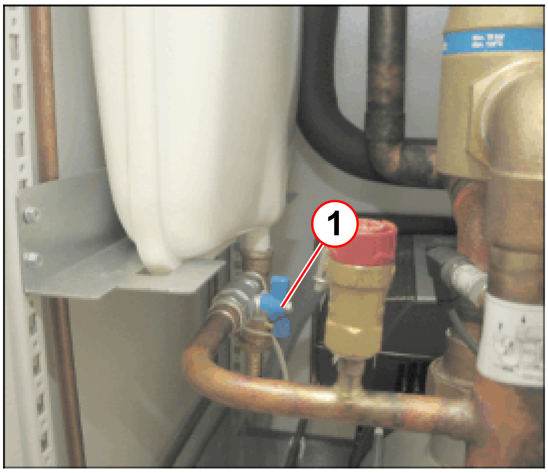
Fig. 33: Air pump

Fig. 34: Autovalve

3.1.5.2 Procedure

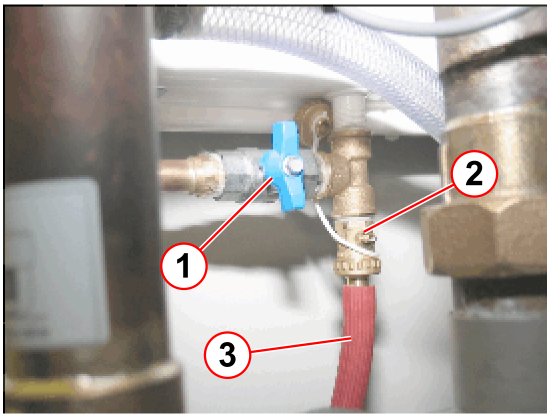
- Switch off the compressor unit.
- Switch off circuit breaker EPC / F102.

Fig. 35: Expansion vessel



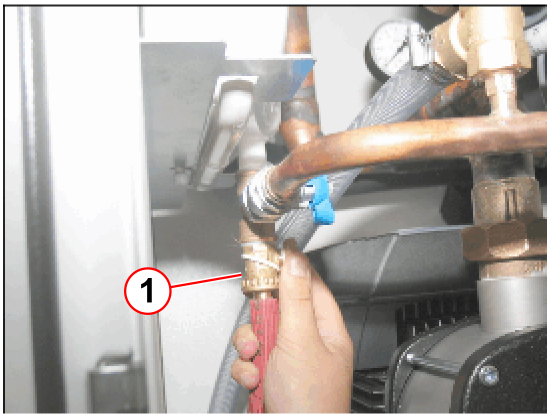
(1) Shutoff valve

Fig. 36: Expansion vessel



(1) Shutoff valve
(2) Drain port
(3) Water hose

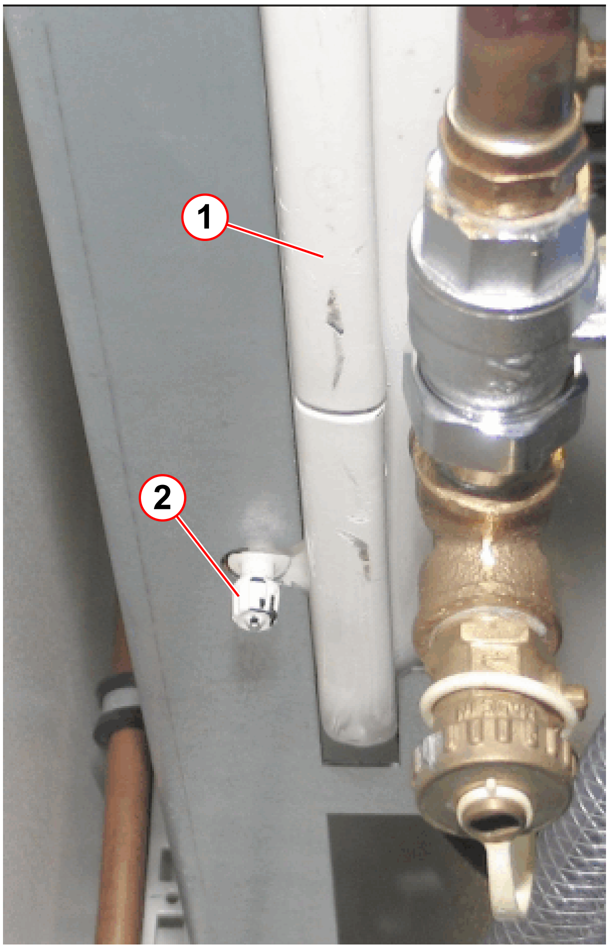
Fig. 37: Expansion vessel



(1) Drain port

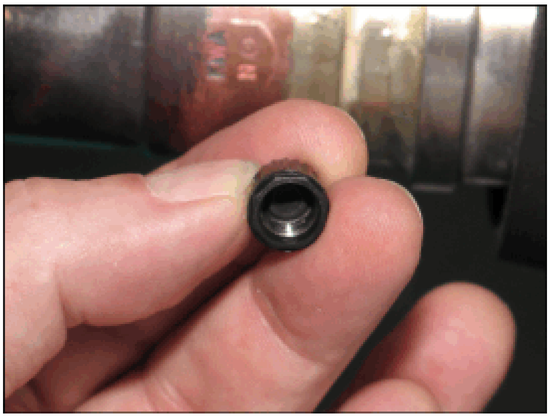
- Close the shut-off valve (→ 1/Fig. 35 Page 32).
- Connect the water hose to the drain port (→ 3/Fig. 36 Page 32).
- Drain the expansion vessel.
  » The volume of expansion vessel is about 8 l of water.
- Remove the cap from the valve (→ Fig. 39 Page 33).

Fig. 38: Expansion vessel



(1) Expansion vessel
(2) Cap

Fig. 39: Cap

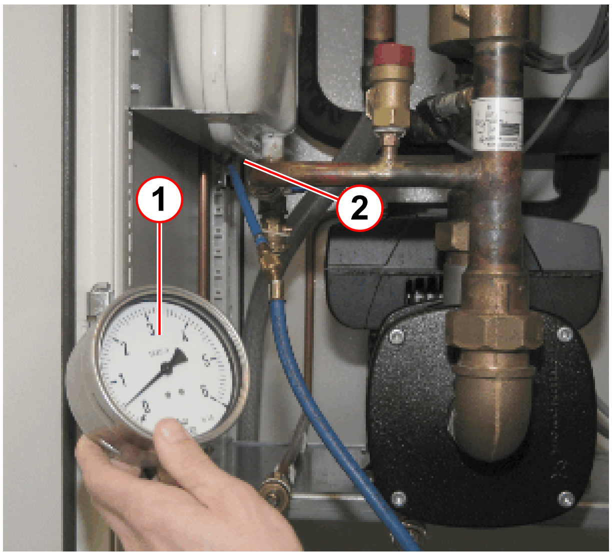


- Connect the measuring device to the fill valve (→ Fig. 41 Page 34), (→ 1, 2/Fig. 40 Page 34).
- Check the pressure.

  » Pressure should be 0.5 bar.

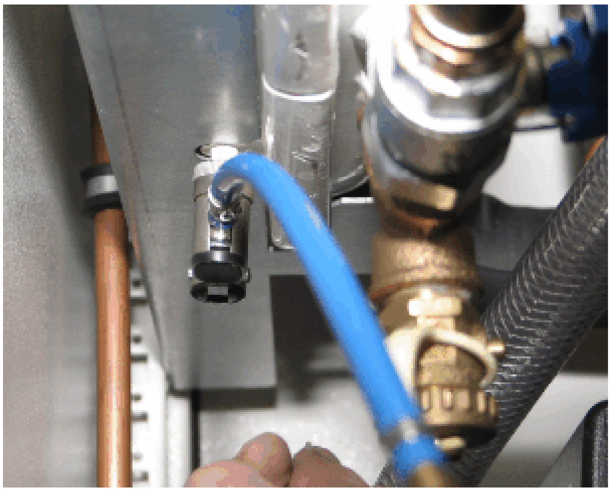
  If not, pressurize the vessel with the air pump.

Fig. 40: Measuring device



(1) Measuring device
(2) Fill valve

Fig. 41: Fill valve

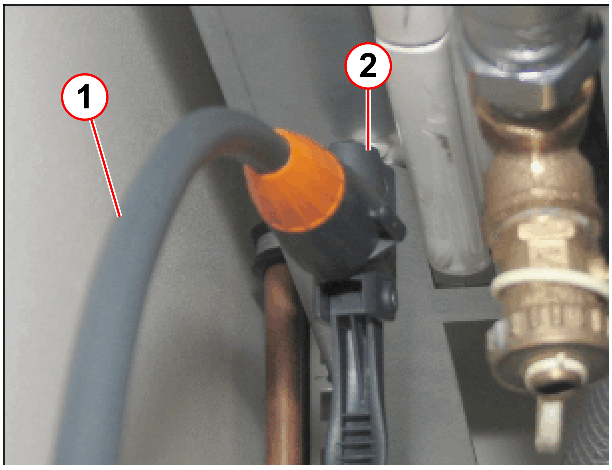


3.1.5.3 Pressurizing the expansion vessel

- Connect the air pump to the fill valve (➔ 1, 2/Fig. 43 Page 35).
- Pressurize the vessel to 0.5 bar.

Fig. 42: Air pump

Fig. 43: Air pump



(1) Air pump
(2) Fill valve

3.1.5.4 Concluding Steps

- Undo the cap on the fill valve.
- Close the drain valve of the expansion vessel.
- Open the shut-off valve of the expansion vessel.
- Fill the secondary water circuit with water, refer to (→ Filling the SEP / Page 22).
- Switch on F102.
  » Water pump will start automatically.
- Start the compressor.
- Check the water pressure after the air is bled.

3.1.6 Conversion list of temperature sensors

Temperature (˚C)	PT 100 (0.4 ohms/˚C)	KTY (~15 ohms/˚C)
	Location: ACS: R1, IFP: Temperature sensor	Location: SEP: B1, B2 (temperature sensor)
0	100 ohms	1,634 ohms
25	110 ohms	2,000 ohms +/- 10 ohms
10	104 ohms	1,774 ohms
20	108 ohms	1,922 ohms
30	112 ohms	2,078 ohms
40	116 ohms	2,040 ohms
50	120 ohms	2,410 ohms

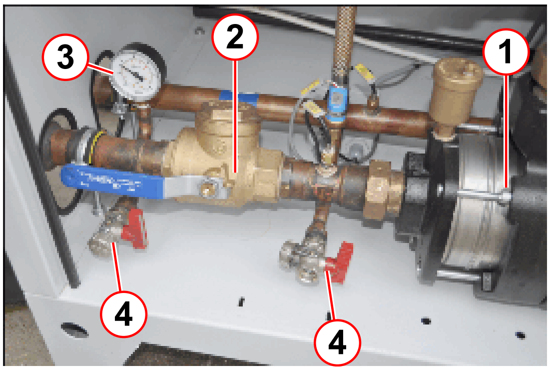
3.2 Chiller

3.2.1 Filling the chiller

> **i** Do not use any connection to a drinking water system.

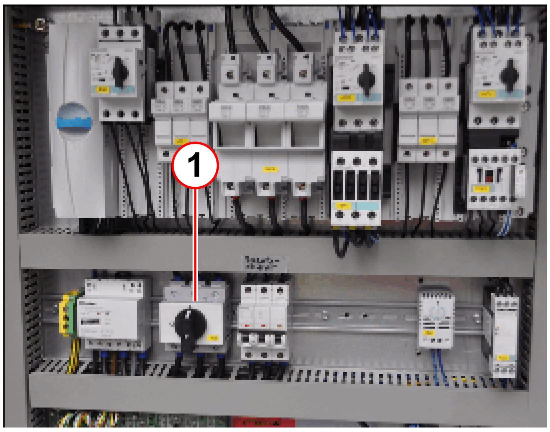
> **i** The water circuit from the chiller to the MR system needs to be filled with 35% to a maximum of 38% vol. ethylene glycol.
>
> 35% to 38% vol. ethylene glycol mixture corresponds to approx. -21 ˚C to -23 ˚C (-5.8 ˚F to -9.4 ˚F). Only ethylene glycol may be used. When starting up the chiller, the chiller start-up personnel should leave 30 liters of ethylene glycol mixture at the system. This reserve supply can be used for refills.

Fig. 44: Chiller



(1) Water pump
(2) Ball valve with strainer
(3) Return water pressure manometer
(4) Drain and fill port

Fig. 45: Chiller electronic cabinet

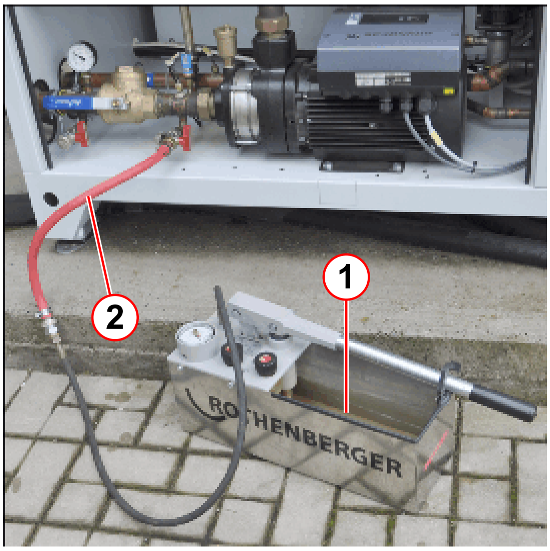


(1) Switch 4Q1

- Switch off the compressor unit.
- Switch off the chiller, using switch 4Q1.

> **i** Before connecting the water hose to the chiller, the hose should be filled with water and not contain air.

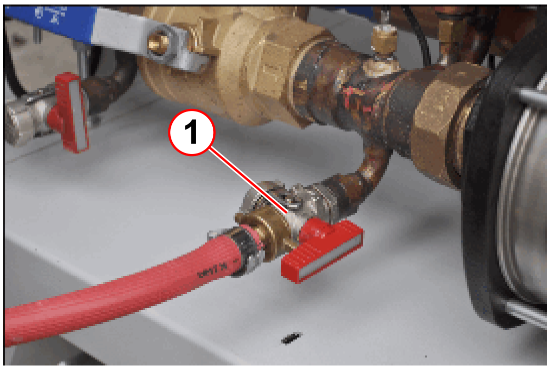
Fig. 46: Chiller



(1) Manual water pump
(2) Water hose

- Fill the water pump with the mixture of ethylene glycol and water.
- Fill the water hose with water before connecting it to the filling port.
- Connect the water hose to the filling port.

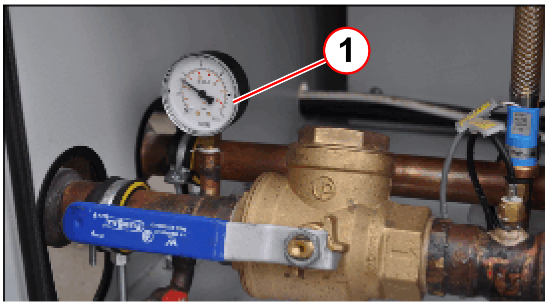
Fig. 47: Chiller



(1) Drain and fill port valve

- Open the valve of the filling port.
- Stop filling when the pressure manometer shows the correct value.

Fig. 48: Chiller



(1) Return water pressure manometer

- Stop filling when the suction pressure is 1.5 bar.

Check the pressure at the IFP manometer

- Chiller **lower** than the IFP or on the **same level**
  - » static water pressure: 1.1 - 1.5bar.

- Chiller **higher** than the IFP panel
  - » static water pressure 1.7 - 1.9bar.

Fig. 49: Chiller

- Close the valve of the filling port.
- Disconnect the water hose.
- Close the filling port with the cap.

- Switch on the chiller.
- Switch on the compressor unit.
- Check for water leaks.

3.2.2 Filling the Auto-Refill Device

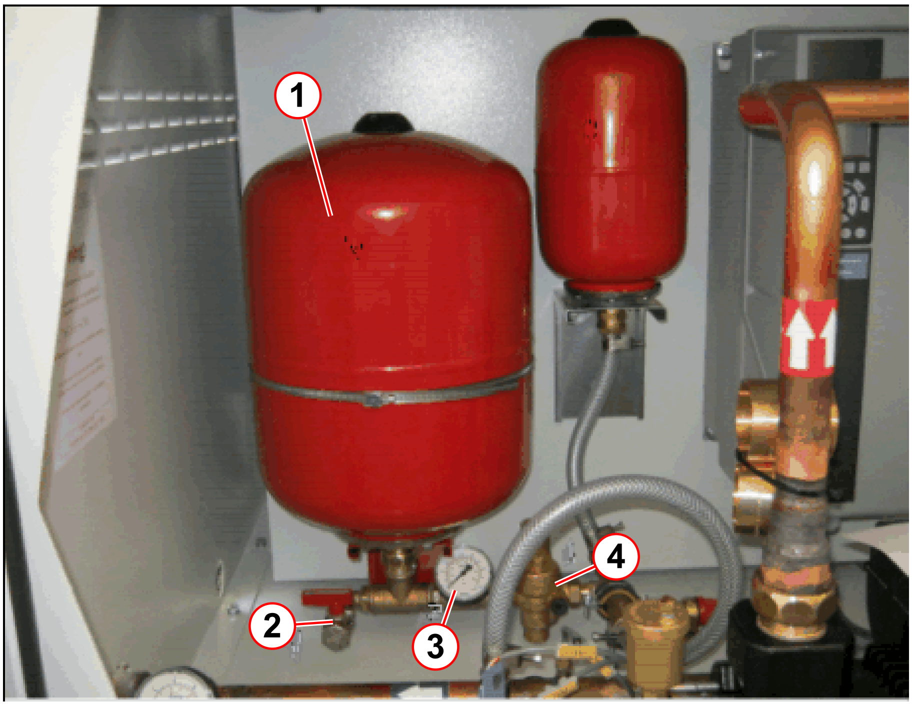
These instructions are only for a chiller with an auto-refill device (second expansion tank).

> The Chillers are equipped with Auto-refill device from serial number:
>
> ECO133: 1153
>
> ECO122: 1076

> The auto-refill device maintains a constant water pressure in the water circuit..
>
> Water is refilled automatically if water loss appears.
>
> Prerequisite: The auto-refill device is filled with antifrogen / water at <6 bar and the pressure regulator is adjusted to 0.6 bar.

> Do not change the setting at the pressure regulator . (→ 4/Fig. 50 Page 40)

Fig. 50: Auto-Refill device



(1) Expansion vessel 25 l
(2) Fill/drain port of expansion vessel
(3) Pressure manometer
(4) Pressure regulator

- Connect the water hose to the fill port of the auto-refill device . (➔ 2/Fig. 50 Page 40)
- Open the fill port slowly.
- Fill the auto-refill device to a range between 5.0 bar and 6.0 bar.

   » Check the pressure via pressure manometer . (➔ 3/Fig. 50 Page 40)

> The expansion vessel needs a filling if the pressure at the pressure gauges is below 1 bar.

3.2.3 Filling at the ACS

An MR system with IFP and ECO chiller can be filled at the ACS.

System with IFP.

Filling the system via the return circuit, low-pressure circuit if the chiller is switched on.

The water circuit from the chiller to the MR system needs to be filled with 35% to 38% ethylene glycol.

35% to 38% vol. ethylene glycol mixture corresponds to approximately -21 ˚C to -23 ˚C (-5.8 ˚F to -9.4 ˚F).

Only ethylene glycol may be used.

When starting up the chiller, the chiller start-up personnel should leave 30 liters of ethylene glycol mixture near the system.

This reserve supply can be used for refills.

Do not use any connection to a drinking water system!

Water can be filled at the ACS with the chiller switched on.

This procedure can be used if water pressure is low or if any water-cooled components were replaced.

3.2.3.1 Required tools

- Water pump (hand pump) (→ Fig. 51 Page 42)
- Filling adapter (→ Fig. 52 Page 42)
- Water hose
- Cooling water with ethylene glycol mixture.

Fig. 51: Water pump

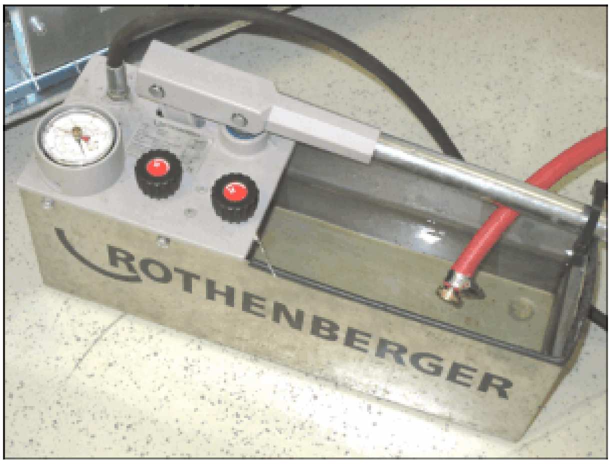


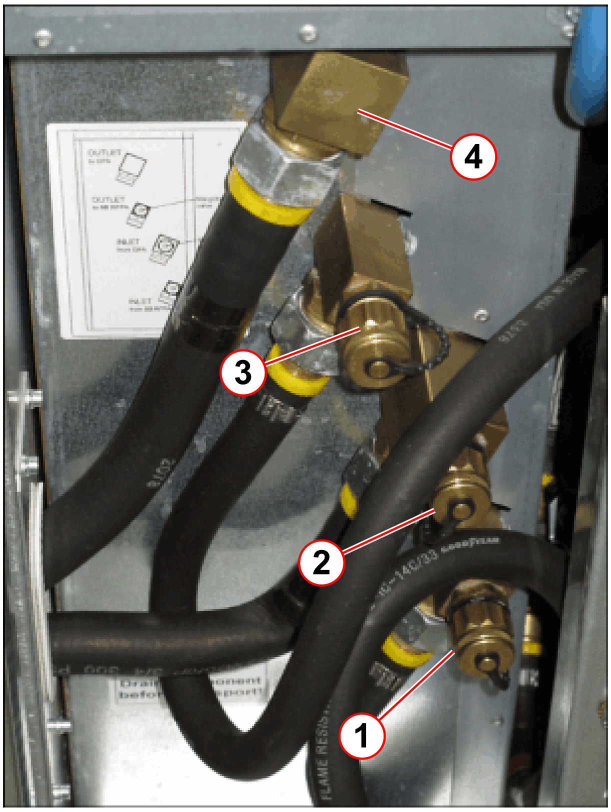
Fig. 52: Filling adapter

3.2.3.2 Procedure

- Fit water hose with filling adapter.
  Filling adapter is delivered with the system.

> Use a filling port of the return pressure circuit. (→ 1 or 2/Fig. 53 Page 43)

Fig. 53: ACS filling port



(1) Return pressure circuit
(2) Return pressure circuit (from GPA)
(3) High pressure circuit
(4) High pressure circuit (to GPA)

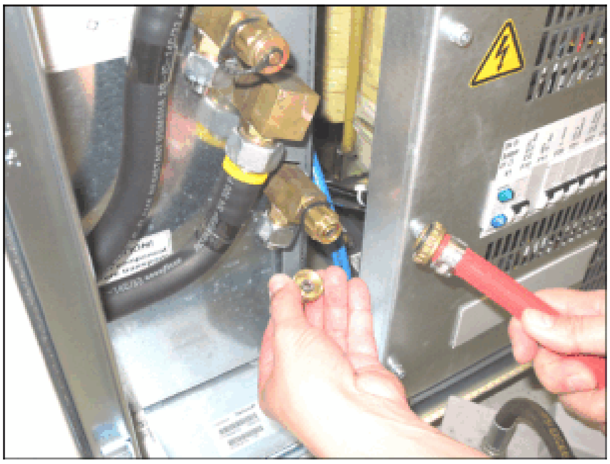
- Before removing the protective cap, check to make sure the valve of the filling port (➔ 1 or 2/Fig. 53 Page 43) is closed.

Fig. 54: Fill and drain port

- Remove the protective cap from the return pressure circuit filling port.

> Before connecting the hose to the ACS, make sure the hose is completely filled with water, and contains no air.
>
> Pump water through the water hose before connecting it.

Fig. 55: ACS filling port



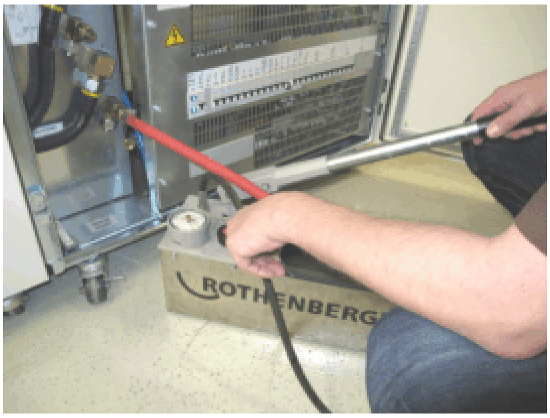
- Fill the pump with ethylene glycol mixture.
- Connect the water hose to the ACS filling port.

Fig. 56: Filling port

(1) Filling port valve

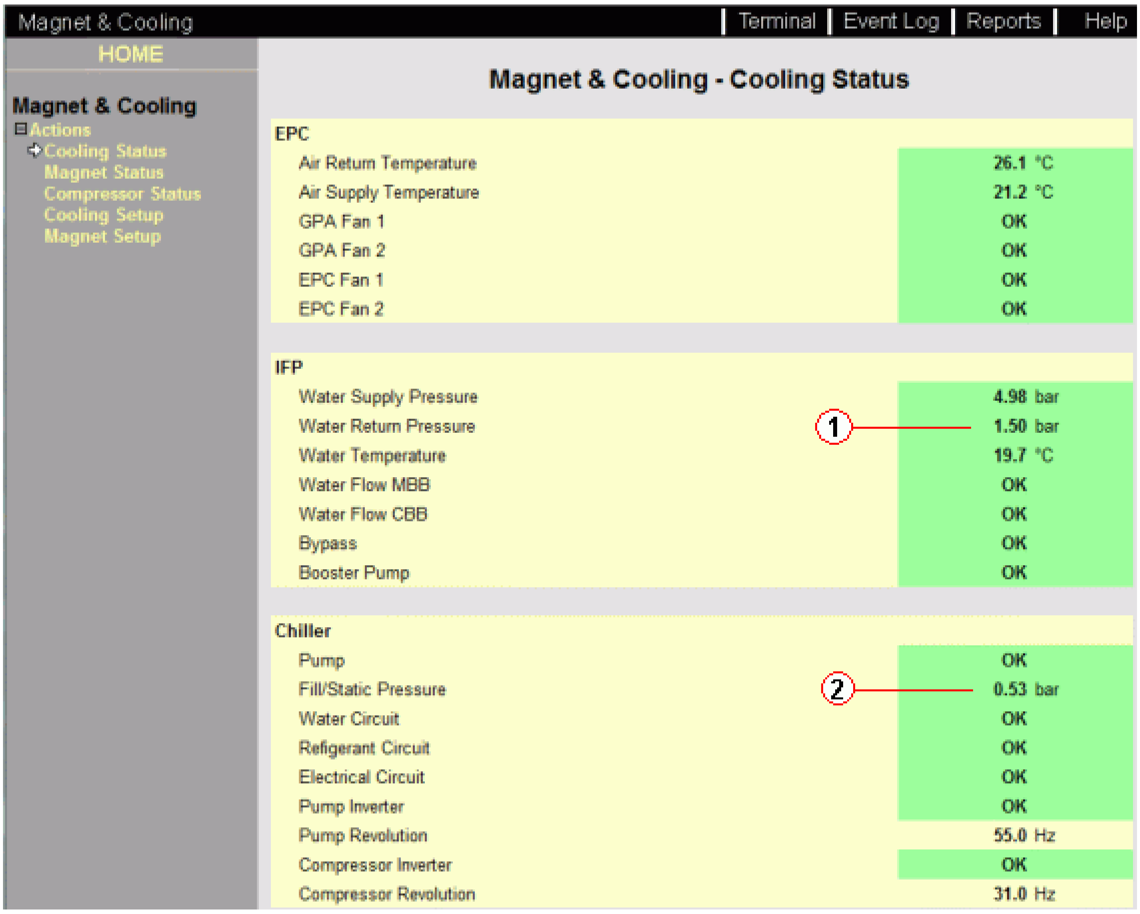
- Open the valve of the filling port.

Fig. 57: Water filling



- Start pumping.
  » Water pressure must be checked via the SeSo platform.
- Close the fill port.
- Open the SeSo platform > "Magnet & Cooling".
  » Check the "IFP water return pressure" (→ 1/Fig. 58 Page 46)
- Refill the "IFP water return pressure" up to 1.5 bar.

Fig. 58: Magnet & cooling



(1) Water return pressure
(2) Fill/static pressure

- Check the “Fill/Static Pressure” of the chiller via the SeSo. (➔ 2/Fig. 58 Page 46)
  - » Increase the water pressure if the fill/static pressure is near the warning value of 0.5 bar.

High difference Chiller - IFP	SeSo IFP: Water return pressure	SeSo Chiller: Fill/static pressure	comment
Chiller 10 m higher	1.5 bar	approx. 0.5 bar	Warning threshold of the chiller fill/static pressure is set to 0.5 bar. » Increase the chiller fill/static pressure to 0.6 bar.
Chiller higher	1.5 bar	<1.5 bar	
Chiller same level	1.5 bar	approx. 1.5 bar	
Chiller lower	1.5 bar	>1.5 bar	
Chiller 30 m lower	1.5 bar	<4.5 bar	

- Disconnect the water hose.
- Place the protective cap on the fill port.

3.2.3.3 Final checks

- Check the filling pressure of the second expansion tank (auto-refill device) in the chiller.
- Refill if fill pressure is low. Refer to (➔ Filling the Auto-Refill Device / Page 39).

3.2.4 Testing the Ethylene Glycol Level

> The specification is 35% vol. to a maximum of 38% vol. of ethylene glycol. Do not use propylene glycol or automotive antifreeze!
>
> The chiller service engineer must perform a check of this concentration during chiller maintenance with a special measurement device.
>
> Antifreeze testers are specified only for certain glycol brands.

3.2.4.1 Required Tools

- A special antifreeze tester. Contact the chiller service department.

3.2.4.2 Procedure

- Open the drain valve on the ACS or on the chiller and drain about 0.5 l ethylene glycol mixture into a bucket.
- Using the temperature scale, check the ethylene glycol level with the ethylene glycol tester.
  - » -21 °C to -23 °C (-5.8 °F to -9.4 °F) corresponds to approx. 35% to 38% vol. of ethylene glycol water mixture.
- Increase the ethylene glycol level if necessary.

> Do not mix different types of glycol with one another, since this can result in flocculation and problems in the chiller.
>
> If an incorrect type of glycol is added, the water circuit will need to be drained completely. Refill the water circuit with a 35% vol. to a maximum of 38% vol. of ethylene glycol water mixture.
>
> Please contact the chiller start-up personnel.
>
> If the chiller is operated without antifreeze, the chiller can be damaged.

3.2.5 Service Mode

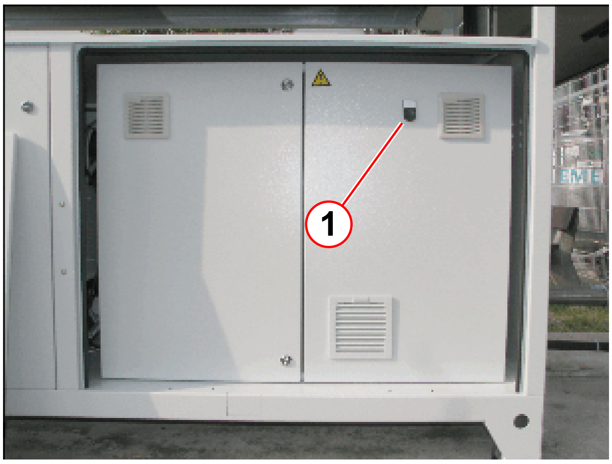
All fault messages to MR System are disabled in Service mode.

3.2.5.1 Procedure

Switch to Service mode

- Press the black reset button for at least 10 seconds.

Fig. 59: Chiller



(1) Reset button

Exit Service mode

- Press the reset button for at least 10 seconds
  » The automatic reset is activated at the latest after 2 hours.

3.2.6 Chiller Reset KSC 215

3.2.6.1 Procedure for KSC 215

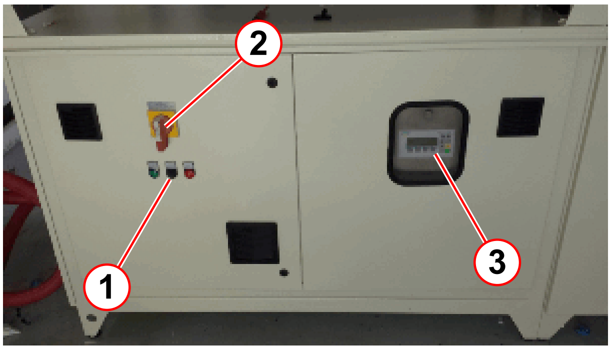
KSC 215- Advanced traditional chiller(s) with logic for the power supply. As soon as the phase monitor recognizes it has overcome the phase lost, phase in-balance or over/under voltage condition, the chiller will restart. If the chiller has pending failures, it will not re-start. The advantage of these chillers is the ability to capture a fault in real-time via Sie-mens Simatic Display. The fault is captured and time stamped. To view older faults in the system we are able to enter the “General Fault” option.

When you encounter a KSC 215 in a failed state, you can do the following:

Simple Procedure

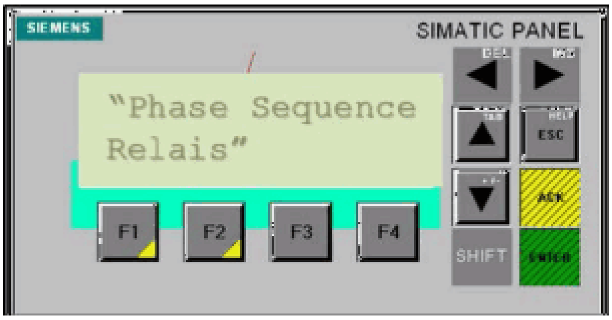
- Observe the Simatic Display through the viewfinder of the chiller panel. A fault will display. (EXP: If faulted on a power issue, "Phase Sequence Relias" will appear.)

Fig. 60: Chiller KSC 215



(1) Reset button
(2) ON / OFF button
(3) Simatic Display

Fig. 61: Simatic display

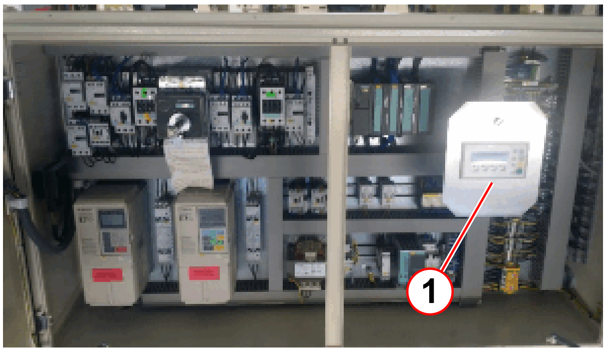


- Press the "Reset button" (→ 1/Fig. 60 Page 49), chiller should run if no other fault condition is present.
- If fault is still present, a complete shutdown of the chiller is recommended. (→ 2/Fig. 60 Page 49)
- If fault is still present, contact the HSC and open a service call.

Advanced Procedure

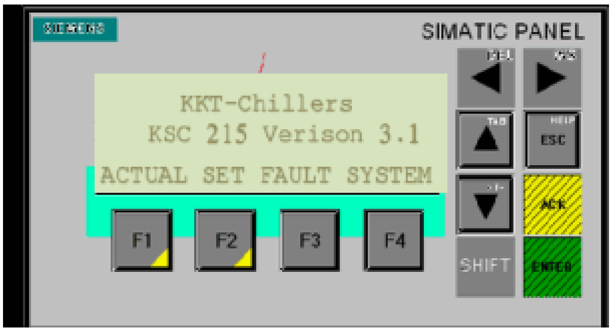
- After pressing the "Reset button" when the fault first appears, we can navigate through the controller (General Fault Option) to find past faults and when they occurred. (This step requires shutdown of the chiller to enter the electrical cabinet via key).

Fig. 62: Chiller Controller



(1) Controller Display

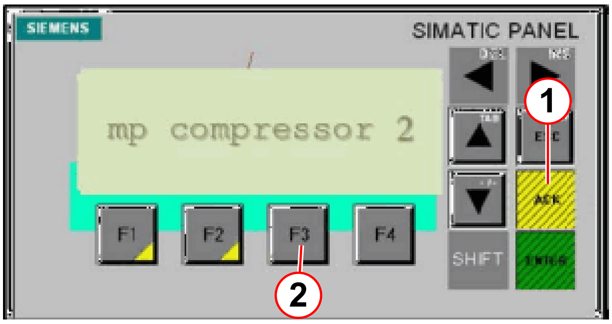
Fig. 63: Actual set fault system



- Switch off the chiller. (→ 2/Fig. 60 Page 49)
- Open the chiller cabinet.

- Press F3, scroll through fault history. (→ 2/Fig. 64 Page 50)
  Record the faults.

Fig. 64: mp compressor 2



(1) ACK
(2) F3

- Press ACK, to acknowledge existing fault. (→ 1/Fig. 64 Page 50)
- Press “Reset button” on the door panel to clear the fault. (→ 1/Fig. 60 Page 49)
- Record the faults and forward to the HSC when opening a “Service Call”. This will help diagnose if the issue is a power failure, etc.

Also, you may want to contact the HSC, and contact Tech Support for additional information. You may be able to receive information when the chiller initially faulted, especially if no fault is present. With the way the chiller communicates with the MR system via Data Cable, we are able to get a well defined failure on the system. (EXP: Compressor Failure, Pump Motor Protection, etc.)

3.2.7 Chiller Reset ECO

3.2.7.1 Procedure for the ECO series

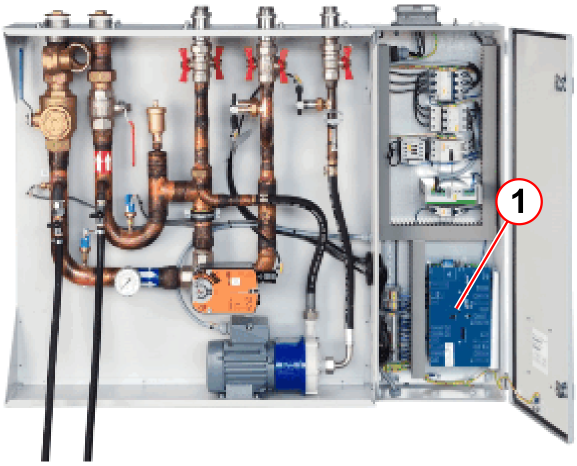
ECO 122/133L- New generation of chiller(s) with logic for the power supply. As soon as the phase monitor recognizes it has overcome the phase lost, phase in-balance or over/under voltage condition, the chiller will restart. If the chiller senses pending failures, it will not restart. The advantages of these chillers are the ability to data log information when the chiller is operating.

When you encounter these chillers in a failed state, you can do the following:

Simple Procedure

- View chiller faults at the IFP board (➔ 1/Fig. 65 Page 51). The system mirrors any fault codes via fiber optic cable with a 1-8 LED.

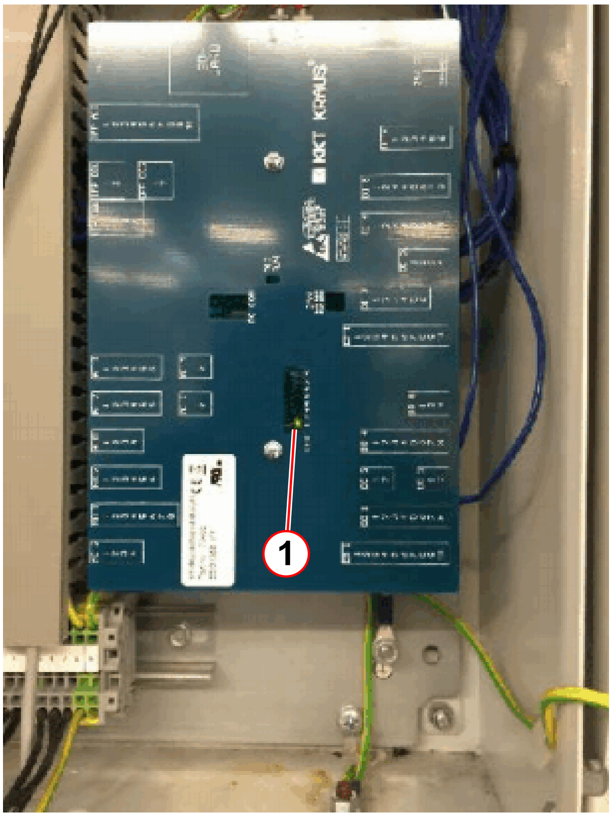
Fig. 65: IFP



(1) IFP Board

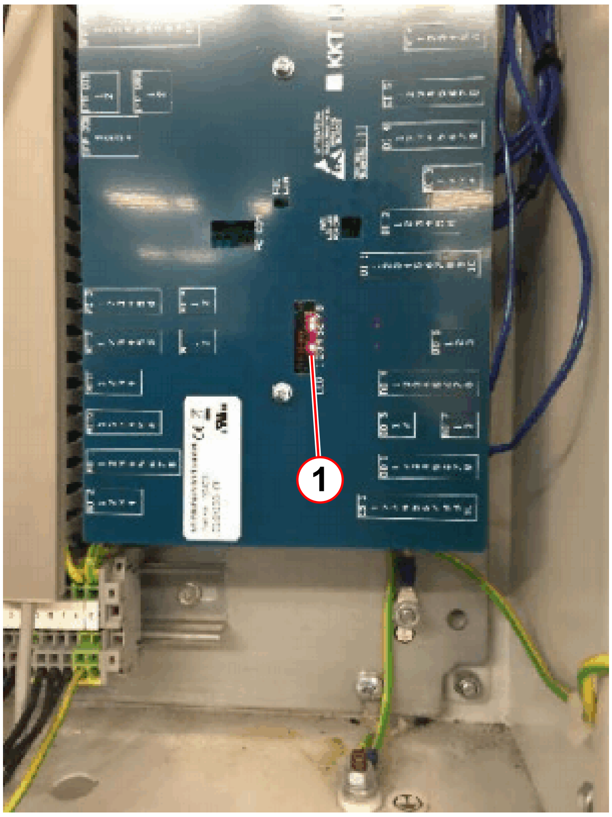
- Record the faults and forward it to HSC. Open a service call. This will help diagnose if the issue is a power failure, etc.

Fig. 66: IFB Board without fault



(1) Green LED's

Fig. 67: IFB Board with fault

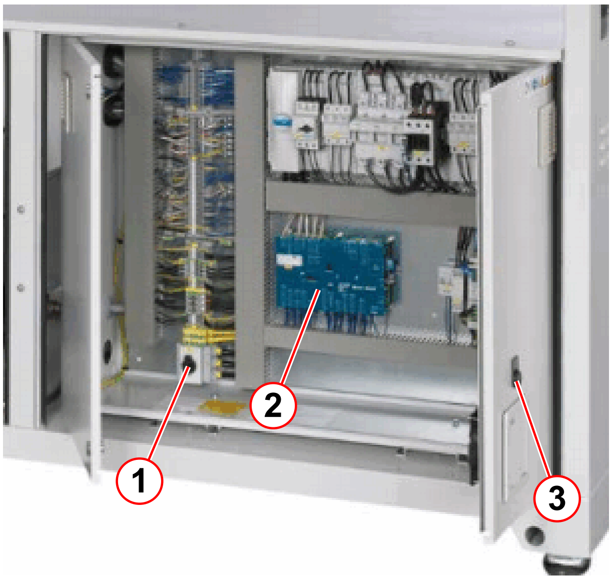


(1) Red LED's

Advanced Procedure

- Access the chiller control cabinet.
- Open the door and record fault codes. (➔ 2/Fig. 68 Page 52)

Fig. 68: ECO Chiller



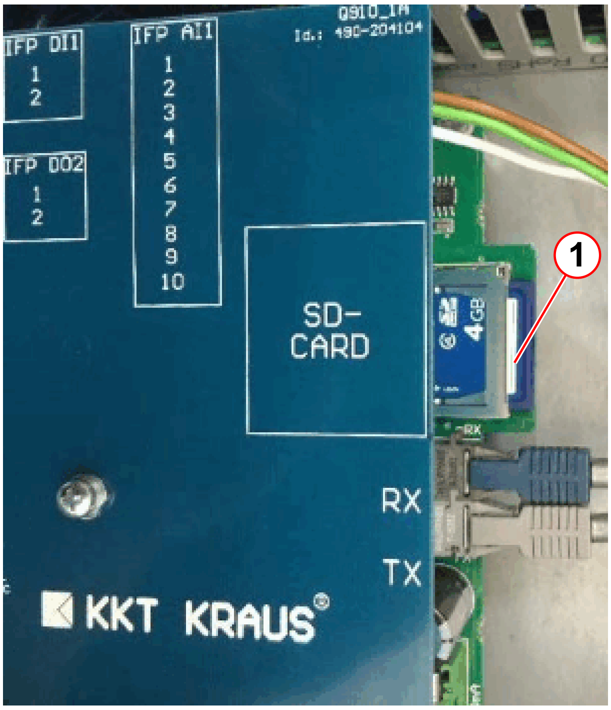
(1) Main switch
(2) Chiller board
(3) Reset button

- Hold reset on the cabinet door for **three** seconds. (➔ 3/Fig. 68 Page 52)
- If cleared, chiller should not indicate any failures.
- If faults are still present, switch off and switch on the chiller.(➔ 1/Fig. 68 Page 52)
- If fault is still present contact the HSC and open a service call..

> i Also, you may want to contact the HSC, and contact Tech Support for additional information. You may be able to receive information when the chiller initially faulted, especially if no fault is present. With the way the chiller communicates with the MR system via fiber optic, we are able to get a well defined failure on the system. (EXP: Compressor Failure, Pump Motor Protection, etc.)

- For additional support, you can send the chiller SD-Card files to Techsupport@kkt-chillersusa.com or mail to the KKT USA Office. (➔ 1/Fig. 69 Page 53)

Fig. 69: Chiller board



(1) SD card

Version 02

- New chapter (→ Testing the expansion vessel / Page 30).

Version 03

- SEP Auto-refill device (→ Filling the auto-refill device. / Page 25) added.
- New chapter (→ Chiller / Page 7).
- Chiller Auto-refill device (→ Filling the Auto-Refill Device / Page 39) added.
- CAN24_7 (→ CAN24_7 Interface / Page 10) added.
- **The following systems were added:**
  - Avanto^{fit}
  - Skyra^{fit}

Version 04

- Prisma, Prisma^{fit} were added.

Version 05

- Chapter (→ Error Codes of the SEP Controller / Page 5)
  - Error codes updated.
- Chiller and IFP run and fault messages added. (→ Chiller / IFP run and fault messages / Page 7)
- Chapter added. (→ Filling at the ACS / Page 40)
- Structural change

Version 06

- New chapter added. (→ Chiller Reset KSC 215 / Page 48)
- New chapter added. (→ Chiller Reset ECO / Page 50)

There are no Hazard IDs in this document.

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